

Information Retrieval

Basic Concepts



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Information Retrieval

- Traditionally dealt with retrieving information about a subject from a collection of natural language documents
 - Locating relevant documents
 - Locating relevant pieces of text within a document
 - Interpreting the content of documents by extracting syntactic and semantic information from document content
 - Information usually ranked according to a degree of relevance
- Early systems were very much concerned with library information systems
 - bibliographic systems
 - document archives
 - special collections of documents for applications such as law, medicine, news services, patents etc.

IR and the Web

- Before Web, IR seen as fairly narrow domain of interest
- The Web
 - Very large, dynamic collection of documents
 - Very large set of non-specialist users
 - Open system that allows many users to publish as well as access information
 - Lot cheaper and easier to have access to various information sources
- IR Revival
 - It would soon be recognised that users required help in finding relevant documents and IR technologies evolved to support web search engines

Keep in Mind

- When presenting the basic concepts of IR you should keep in mind how you find information – whether using a library, a particular book, your notes or the web
 - *How do you find books in a library about a particular topic?*
 - *How do you locate information within a book about a particular topic?*
 - *How do you find web pages about a particular topic?*
 - *How do you find images of interest on the web?*
 - *When revising for an exam how do you find information about a particular topic?*

Information Retrieval ...

- Nowadays covers not only textual documents, but also other forms of unstructured data such as
 - **Image**
 - *find images with a castle, a highland cow and blue sky*
 - *find images similar to my image of a sunset*
 - **Video**
 - *find clips showing goals scored in football broadcasts*
 - **Audio**
 - *find news broadcasts about natural disasters in Switzerland*
- May also have to deal with different languages
 - *finding Swiss radio news reports about topics where queries may be expressed in different languages and reports may be in different languages*

IR Systems and DB Systems

- **DB Systems**
 - Data has well-defined structure and semantics
 - Data retrieval language aimed at retrieving all objects that satisfy a clearly defined condition
- **IR Systems**
 - Deals with unstructured, or semi-structured, data where the content could be semantically ambiguous
 - Information retrieval language aimed at finding relevant information about a subject, often expressed as a set of keywords

IR Related Areas

- Database Management
- Library and Information Science
- Artificial Intelligence
 - formalisation for representing knowledge and queries
- Natural Language Processing
 - analysis of texts to allow retrieval based on meaning rather than simply keywords
- Machine Learning
 - automated classification of documents

Information Retrieval

- In this block of three lectures, we will discuss
 - the basic concepts of information retrieval
 - information retrieval models
 - retrieval evaluation
 - document pre-processing
 - indexing and searching for documents within a document collection
 - searching the web
 - indexing and searching for web documents
 - *recall discussions on web agents which looked at how to extract "structured information" from web documents*
 - multimedia information retrieval
 - indexing and searching for images

Document Search

- Overview of Techniques for Information Retrieval
 - Search often based on Keywords with Boolean Operators
 - Exclude Irrelevant Words from Documents
 - Reduce Similar Words to Single Term using Stemming
 - Allocate Weightings to Terms based on how useful they are likely to be in deciding whether or not a document is relevant
 - use some sort of ranking algorithm to order documents according to how relevant they are likely to be

Information Retrieval Systems

Boolean Query

term1 AND term2 OR term3

ranked
relevant
documents

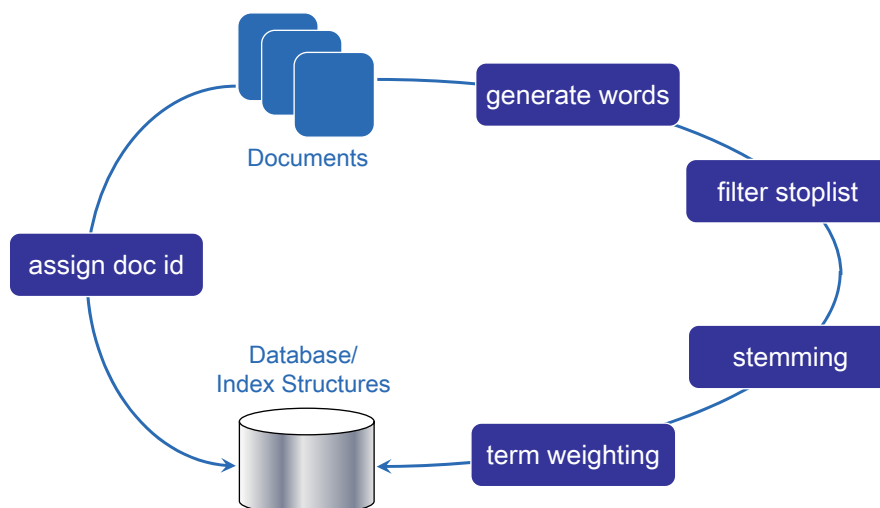


document
collection

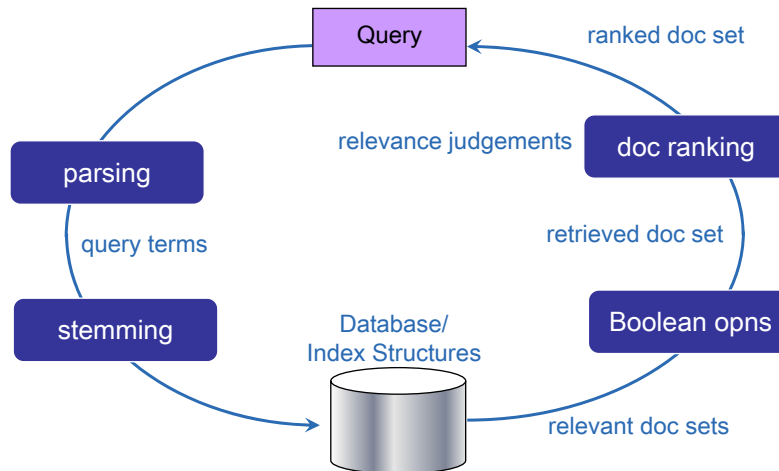
Operations on Terms

- Truncation
 - use of wildcards e.g. 'comp*'
- Stemming
- Stoplist
 - words with no indexing value e.g. 'a', 'the', 'of',
- Weighting
 - significance measure

Document Entry



Query Processing



Document Preprocessing

- 5 main text operations
 - Lexical analysis of text for handling of digits, hyphens, punctuation marks, case of letters
 - Elimination of stopwords to filter out words with very low discrimination values
 - Stemming of remaining words to handle syntactic variations of words by removing prefixes and suffixes
 - Selection of index terms which is usually related to syntactic nature of word (noun, adjective, verb etc)
 - Construction of term categorisation structures such as thesaurus

Lexical Analysis of Text

- Basically processing of converting stream of characters into a stream of words, but certain cases require care
 - Numbers generally not good index terms since they have little meaning without surrounding context, but could be useful when mixed within a word
 - e.g. 800BC, O2

If dates etc. used as index terms, they should be normalised.

- Hyphens not used consistently and one way to deal with them is to break up hyphenated words into separate words
 - e.g. four-wheel-drive
 - e.g. so how would we deal with "sub-type" or "mark-up"?

Lexical Analysis of Text ...

- Generally punctuation removed as part of processing, but in some cases may be significant
 - e.g. "e.g.", "e.g" and "eg" match
 - e.g. how should we handle text containing "index.html"?
- Generally case of letters not important and all text converted to either upper or lowercase.
 - "WEB", "Web" and "web" all match
 - But "Beat" can be seen to be different from "beat" so how should we deal with this?

Lexical Analysis of Text ...

- Lexical analysis is straightforward to implement but care needs to be taken about the rules used as could have a significant effect on document retrieval.
- Many of the cases we have presented may be dealt with through general rules together with rules to handle special cases.
- To avoid such issues, some systems, notably web search engines, do not use such text processing operations and instead index the full text.

Elimination of Stopwords

- Words which occur frequently and have little semantic meaning are poor discriminators among documents and can therefore be filtered out
 - Articles: the, a, an
 - Prepositions: of, on, at, ...
 - Conjunctions: and, or,
- May also be extended to remove some verbs, adverbs and adjectives
 - Some lists of appropriate stopwords have been published
- Also has the benefit of greatly reducing the size of the indexing structure e.g. 40% reduction can result due to elimination of stopwords.
- But again can reduce recall e.g. what happens when we want to retrieve documents containing "to be or not to be"?

Stemming

- User queries often specify variants of word as it appears in text
 - e.g. computer, computing, computed, ...
 - to improve recall we would like these terms to produce a match and one way of doing this is to reduce words to their grammatical stem
- A stem is the portion of a word left after removal of suffixes and prefixes introduced for tenses, plurality, modality etc. However, not always so simple ...
 - trees → tree, skies → sky
 - computing → ?
- Clearly rules are very much language specific

Table Look-up Stemmers

Term	Stem
engineering	engineer
engineered	engineer
engineer	engineer

- Fast but High Storage Overhead
- Such Data not available for all Languages
- Problems of Domain Dependent Data

Affix Removal Stemmers

- Remove Suffixes and/or Prefixes
- Possibly Transform Resultant Stem
- Example

remove plurals from terms

if term ends in “ies” but not “eies” or “aies”

then “ies” -> “y”

if term ends in “es” but not “aes”, “ees” or “oes”

then “es” -> “e”

if term ends in “s” but not “us” or “ss”

then “s” -> NULL

Affix Removal Stemmers

- Iterative Longest Match (Lovins 1968)
 - remove longest possible string of characters according to rule set until no more characters can be removed
 - skies -> ski
- Recode
 - use context sensitive information s.t. ski -> sky
- Partial Match
 - use only n initial characters in matching e.g ski # sky

Porter Stemming Algorithm

- Set of Condition/Action Rules with Conditions on Stem

measure	based on alternate vowel-consonant sequences
*<X>	stem ends with given letter X
V	stem contains a vowel
*d	stem ends in a double consonant
*o	stem ends with consonant-vowel-consonant where final consonant not w, x or y

Measure

- Vowels : a, e, i, o, u and y if preceded by a consonant
- Measure m given by

$$[C] (VC)^m [V]$$

where V is a seq of vowels, C a seq of consonants

Measure

Examples

m=0

TR, EE, TREE, Y, BY

m=1

TROUBLE, OATS, TREES, IVY

m=2

TROUBLES, PRIVATE

Example Rules

Condition	Suffix	Replacement	Example
m>0	ational	ate	relational -> relate
m>0	tional	tion	conditional -> condition
m>0	izer	ize	digitizer -> digitize
m> 0	ation	ate	predication -> predicate
m>0	ator	ate	predication -> predicate
m>0	alism	al	feudalism -> feudal
m>0	entli	ent	differentli -> different
m>0	fulness	ful	hopefulness -> hopeful

Selection of Index Terms

- Choices
 - full text representation – all words in text used as index terms
 - manual selection of terms by experts – often used in bibliographic systems
 - automatic selection of terms – often based on noun groups
 - Argued that most of the semantics of a sentence carried by nouns
 - Nouns commonly grouped e.g. computer science, so nearby nouns grouped into single indexing component
 - Noun groups may be defined in terms of syntactic distance – threshold of number of words between nouns

A FRAMEWORK FOR CROSS-MEDIA INFORMATION MANAGEMENT

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ABSTRACT

We present an information server architecture, called iServer, that enables cross-media linking based on an object-oriented hypermedia model. The iServer mixed-media platform can easily be extended to support new types of digital or physical multimedia resources. It provides core functionality for link and user management whereas link authoring is handled by plug-ins for specific resource types. Different forms of collaborative cross-media link authoring and link sharing are supported by a distributed iServer implementation based on peer-to-peer technologies.

KEY WORDS

Cross-media information management, multimedia integration, hypermedia model, link server

1 Introduction

The ubiquitous computing community are interested in developing information environments that span digital and physical spaces and offer context-dependent access to information through a range of media and devices. To do this, they are applying hypermedia concepts to structure and link information spaces in a flexible way while en-

gies to support flexible means of sharing dynamically authored links among communities of users. Instead of having a central server, each user becomes an author of his personal link space managed by a local iServer instance and, at the same time, contributes as a publisher to the community's link knowledge.

iServer is based on an object metamodel and object-oriented database technologies. As we will show in this paper, our approach led naturally to a very general and flexible link model that supports various concepts of current hypermedia systems, but also caters for cross-media linking and extensibility for new resource types. The iServer metamodel is a generalisation of an information model that was initially developed within the European research project Paper++ [1] for linking printed and digital media.

We begin in Sect. 2 by motivating the design and implementation of iServer, a general and flexible cross-media information platform. Section 3 then introduces the core concepts of the iServer link model. In Sect. 4, we describe how iServer can be applied to link semantically rich information resources. Collaborative link authoring and link sharing together with several applications that have been implemented on the iServer platform are discussed in Sect. 5. Concluding remarks are given in Sect. 6.

EdFest: Mobile Information System for Visitors to the Edinburgh Festivals

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ABSTRACT

We describe a mobile tourist information system for visitors to the Edinburgh international arts festivals that was demonstrated and tested at the 2004 festivals. The interface to the system is based on a digitally augmented paper brochure and voice dialogue. We present the overall architecture and describe how existing components for multi-channel web publishing, content awareness, and cross-media linking were extended and integrated to provide the necessary functionality, detailing the operation of the Client Controller. We report on our experiences in terms of both technological issues and user studies.

Categories and Subject Descriptors

H.3.4 [Information Storage and Retrieval]: Systems and Software—current awareness systems, distributed systems; H.5.2 [Information Interfaces and Presentation]: User Interfaces—input devices and strategies, voice I/O

General Terms

Design, Experimentation

Keywords

Augmented paper, content management, context-awareness, cross-media information management, mobile information system, multi-modal interface, user trials

1. INTRODUCTION

Mobile information systems require platforms that not only

If we consider the domain of tourism which has been a focus of a large number of research projects in mobile information systems, many of these requirements are not met in terms of the devices and forms of interaction provided. Tourism is generally a social activity and part of the enjoyment is planning activities together with family and friends. PDA screens are small and difficult to read outdoors and by more than one person at a time. The small screen size limits the amount of information that can be viewed at one time and does not support the actions of comparing and combining information which is often what tourists want to do. Some researchers have therefore experimented with tablet PCs to provide better functionality [6], but clearly these further restrict mobility as they are much heavier and require the use of both hands. Various other options such as head-mounted displays, digitally augmented paper and audio have received less attention but clearly are all worthy of consideration and experimentation, either alone or in combination.

Our interest is to not only investigate new forms of information access and interaction in mobile environments, but also to develop information system platforms capable of supporting them and the experimentation process. To achieve this, we have embarked on a multi-stage project that involves developing a mobile system to support visitors to the Edinburgh festivals. The first stage involved developing and performing user trials with a system where the interaction was based on digitally augmented paper and audio. The system had to be developed within a four month period in order that it could be tested at the 2004 festivals. This was achieved through the interaction of researchers

Use of Thesauri

- To ensure good recall, must be able to match terms specified in user queries to synonyms and also to generalisations and specialisations of terms
- A thesaurus provides the required information and some normalisation of vocabulary and concept structure
- Roget described a thesaurus as
 - a classification scheme composed of words and phrases whose organisation aims at facilitating the expression of ideas in written text
- Example entry from Roget's thesaurus

cowardly adjective
Ignobly lacking in courage: *cowardly turncoats*.
Syns: chicken (slang), chicken-hearted, craven, dastardly, faint-hearted, gutless, lily-livered, pusillanimous, unmanly, yellow (slang), yellow-bellied (slang).

Use of Thesauri ...

- A thesaurus may be general or domain specific
 - Thesaurus of Engineering and Scientific Terms
- A thesaurus may be used in several ways:
 - To help users formulate queries
 - Modification of queries by the system
 - To select index terms
- A further challenge is thesaurus construction and how to do this automatically

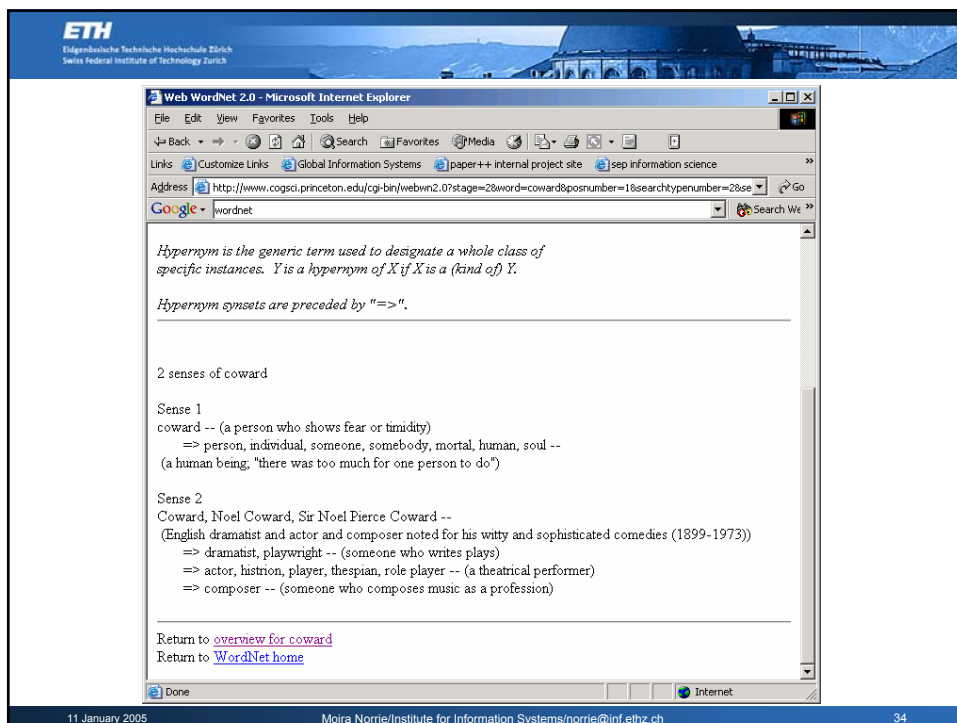
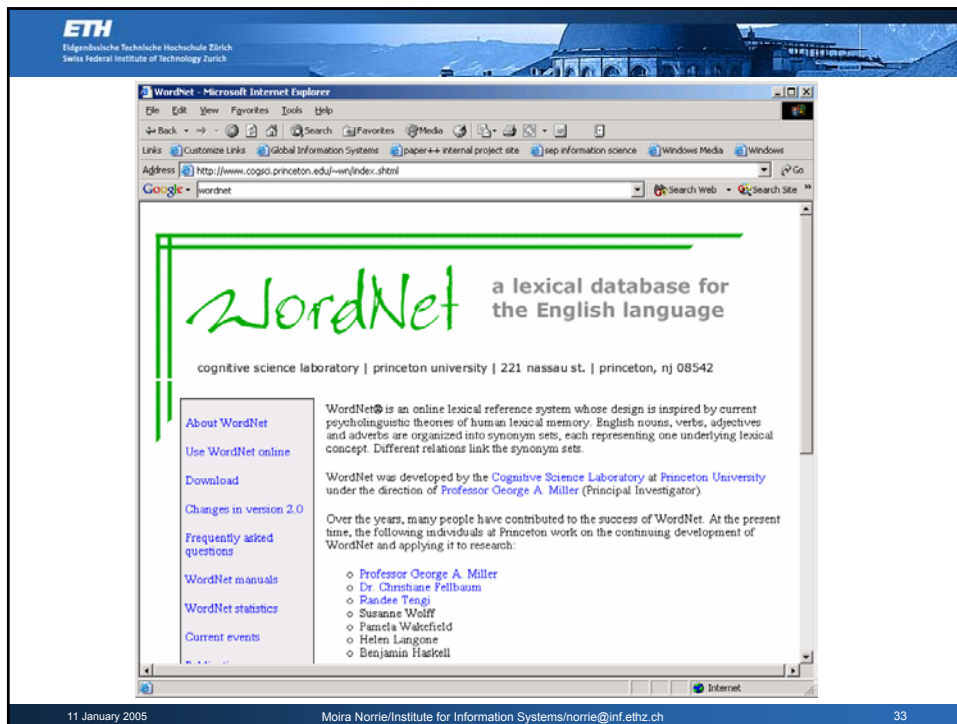
INSPEC Thesaurus

- Domain : physics, electrical eng, computers
- Organised as a set of hierarchies
 - TT : root or top term
 - NT : more specific term
 - BT : more general term
 - RT : related term
 - UF/USE : chosen form
 - CC/FC : INSPEC classification

INSPEC Thesaurus

Example

computer-aided instruction
see also education
UF teaching machines
BT educational computing
TT computer applications
RT education
teaching
CC C7810C
FC c7810Cf



Information Retrieval Models

- Three classic models
 - **Boolean**
 - Documents and queries represented as sets of index terms
 - standard operations on sets
 - **Vector**
 - Documents and queries represented as vectors in a 3-dimensional space
 - standard linear algebra operations on vectors
 - **Probabilistic**
 - Modelling of documents and queries based on probability theory
- Later we will describe the basic mechanisms of the Boolean and Vector models

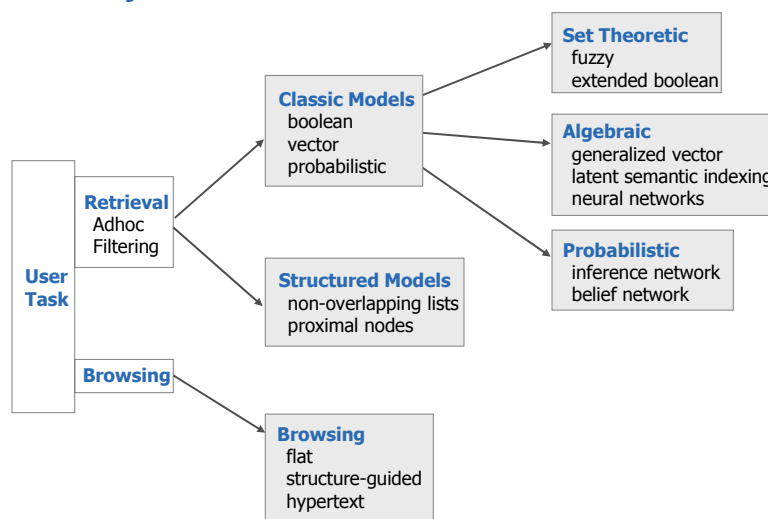
Information Retrieval Models ...

- Many variants of basic 3 models have been proposed
- Logical view of documents may also vary
 - **Index terms**
 - a set of index terms are used to describe the contents of a document
 - **Full text**
 - rather than trying to capture the semantic content of a document in terms of a set of index terms, the full document text is used
 - **Full text plus structure**
 - structural models refer to the structure that is normally present in written text and not just the content

Information Retrieval Models

- In addition to different models for retrieval, there are also different models for document browsing
 - Flat
 - Structure guided
 - Hypertext
- *When do you use these different styles of browsing?*
- *How do they relate to your browsing of web documents?*

Taxonomy of IR Models



Ad hoc and Filtering Retrieval

- Ad hoc Retrieval
 - In conventional IR systems, the document collection is relatively stable and users submit new queries to the system
 - Libraries of documents
 - The Web
- Filtering Retrieval
 - Queries remain relatively static and new documents which come into the system are filtered
 - News services, stock market data, ...
 - User profiles form basis of filtering. Profiles may be keyword-based with users specifying these manually or system generating them based on relevance feedback
 - See earlier lecture on web information agents

IR Model

- An information retrieval model is a quadruple $[D, Q, F, R(q_i, d_j)]$ where
 - D is a set of composed logical views (or representations) for the documents in the collection
 - Q is a set composed of logical views (or representations) for the information needs called queries
 - F is a framework for modelling document representations, queries and their relationships
 - $R(q_i, d_j)$ is a ranking function which associates a real number with a query $q_i \in Q$ and a document representation $d_j \in D$. Such a ranking defines an ordering among documents with regard to the query q_i

Index Terms

- In classic models, each document is described by a set of index terms that describe what the document is about
 - May extract certain nouns which generally are better at determining semantics of document content than adjectives, adverbs etc
 - Alternatively may consider all distinct words of document as index terms – *full text logical document view*
(Approach adopted in some Web search engines as to be discussed later)
- Deciding on suitable index terms not trivial
 - Consider words appearing in collection of documents
 - Words contained in all documents not as useful as words which appear only in small set of documents
 - e.g. in a collection of documents on databases, the term "database" would not be very useful

Weighted Index Terms

- Let k_i be an index term, d_j be a document then we associate weight $w_{i,j}$ with (k_i, d_j) to quantify the importance of k_i for describing the contents of d_j
- Let $K = \{k_1, k_2, \dots, k_n\}$ be the set of index terms for a collection of documents $D = \{d_1, d_2, \dots, d_m\}$ and assume weights $w_{i,j}$ associated with (k_i, d_j) for $i=1, \dots, n$ and $j=1, \dots, m$ such that $w_{i,j} = 0$ if k_i does not appear in d_j
- Then for each document d_j in D there is an index term vector $\text{vec}(d_j) = (w_{1,j}, w_{2,j}, \dots, w_{n,j})$
- Further we introduce a function g_i that returns the weight associated with index term k_i in any of the index term vectors such that $g_i(\text{vec}(d_j)) = w_{i,j}$

Weighted Index Terms ...

- Usually assume that index term weights mutually independent
 - e.g. knowing $w_{i,j}$ tells us nothing about $w_{i+1,j}$

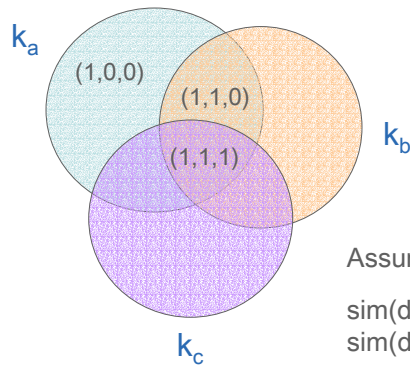
although this is a simplification since occurrences of index terms not uncorrelated

- e.g. the terms *information* and *retrieval* may be index terms for a collection of documents – but they will often occur together
- Assuming mutual independence greatly simplifies the task of computing index term weights and ranking, and research is only recently beginning to show how such correlations can be used to advantage

Boolean Model

- Index terms are either present or not
 - $w_{i,j} \in \{0,1\}$
- Queries are Boolean expressions consisting of index terms and logical connectives $\wedge, \vee$ and $\neg$
- Example
 - Query $q = k_a \wedge (k_b \vee \neg k_c)$
 - Rewritten in disjunctive normal form as
$$\text{vec}(q_{\text{dnf}}) = (1,1,1) \vee (1,1,0) \vee (1,0,0)$$
where each of the binary weighted vectors associated with the tuple (k_a, k_b, k_c)

Boolean Model ...



$$q = k_a \wedge (k_b \vee \neg k_c)$$

Assume q_{cc} is a conjunctive component of q_{dnf}

$$\text{sim}(d_j, q) = 1 \text{ if } \forall k_i, g_i(\text{vec}(d_j)) = g_i(\text{vec}(q_{cc}))$$

$$\text{sim}(d_j, q) = 0 \text{ otherwise}$$

If $\text{sim}(d_j, q) = 1$ then d_j is considered to be relevant to the query q

Boolean Model

- Simple and used in many early commercial bibliographic systems
- Suffers from major drawbacks
 - Retrieval strategy based on binary decision criteria – documents either relevant or not relevant – no notion of partial matching
 - No ranking of documents
 - Not always easy to translate an information need into a Boolean expression and usually users express only simple queries in this way
- Due to drawbacks, the Boolean model frequently returns either too few or too many documents in response to the user query

Vector Model

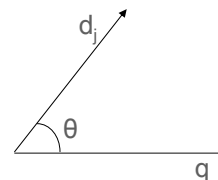
- Allows partial patching by assigning non-binary weights to index terms and documents
- Compute a degree of similarity for each document in response to a given query
- Documents sorted in decreasing order of degree of similarity
- Assume set of index terms is $K = \{k_1, k_2, \dots, k_n\}$

Then both document d_j and query q are represented as n -dimensional vectors of form $[w_1, w_2, \dots, w_n]$ where w_i for $i=1, \dots, n$ is the weight of the index term k_i for the document or query

Degree of Similarity

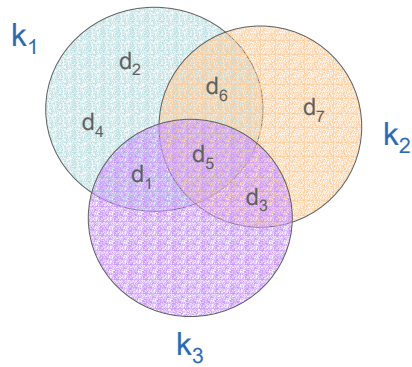
- Assume $\text{vec}(d_j)$ and $\text{vec}(q)$ are the weighted vectors for document d_j and query q , respectively
- Then the degree of similarity is a measure of the correlation between $\text{vec}(d_j)$ and $\text{vec}(q)$ and can be quantified by the cosine of the angle between the two vectors

$$\text{sim}(d_j, q) = \frac{\text{vec}(d_j) \cdot \text{vec}(q)}{|\text{vec}(d_j)| \times |\text{vec}(q)|}$$



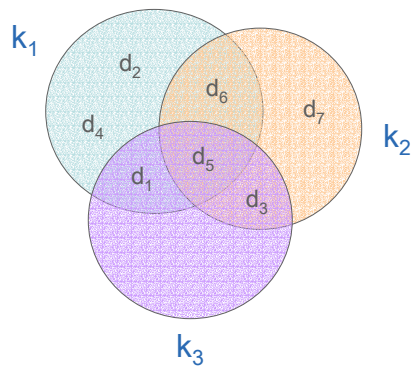
$\cos \theta$ adopted as $\text{sim}(d_j, q)$

Example 1



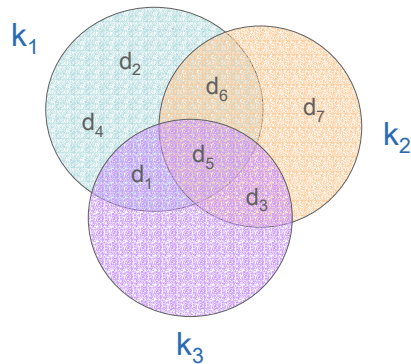
	k_1	k_2	k_3	$q \cdot d_j$
d_1	1	0	1	2
d_2	1	0	0	1
d_3	0	1	1	2
d_4	1	0	0	1
d_5	1	1	1	3
d_6	1	1	0	2
d_7	0	1	0	1
q	1	1	1	

Example 2



	k_1	k_2	k_3	$q \cdot d_j$
d_1	1	0	1	4
d_2	1	0	0	1
d_3	0	1	1	5
d_4	1	0	0	1
d_5	1	1	1	6
d_6	1	1	0	3
d_7	0	1	0	2
q	1	2	3	

Example 3



	k_1	k_2	k_3	$q \cdot d_j$
d_1	2	0	1	5
d_2	1	0	0	1
d_3	0	1	3	11
d_4	2	0	0	2
d_5	1	2	4	17
d_6	1	2	0	5
d_7	0	5	0	10
q	1	2	3	

Index Term Weights

- How are the index term weights determined?
 - Many different methods proposed
 - Here we present only the main idea behind the most effective methods which are based on clustering techniques
- IR as a clustering problem
 - collection C of documents
 - user query q is a vague specification of some subset A of C
 - problem is to determine which documents are in A and which are not
 - clustering problem
 - what features provide a good characterisation of objects in A ?
 - what features discriminate well between the objects in A and the rest of the objects in C ?

Calculation of Term Weights

- Two effects must be taken into account
 - quantification of intra-document contents
 - the term frequency within a document (**tf factor**)
 - a measure of how well the term describes the document content
 - quantification of inter-document contents
 - measure of how discriminating terms are across a collection of documents
 - the more documents in which a term appears the less useful it is in distinguishing relevant from irrelevant documents
 - the inverse document frequency (**idf factor**)
- Calculation of Weights for term i in document j

$$w_{ij} = \text{tf}(i,j) * \text{idf}(i)$$

TF and IDF Factors

- Let
 - N be total number of documents in a collection
 - n_i be number of documents containing k_i
 - $\text{freq}(i,j)$ be the number of occurrences of k_i in document d_j
- The normalized frequency factor (tf factor) is given by
 - $f(i,j) = \text{freq}(i,j) / \max_l(\text{freq}(l,j))$
 - max computed over all terms in d_j
- The idf factor for a term k_i is computed as
 - $\text{idf}(i) = \log(N/n_i)$
 - log used to make values of tf and idf comparable

Development of IR Systems

- The development of information retrieval systems often consists of the following steps;
 - Acquisition of test document set
 - Construction of test query set
 - Development of retrieval model
 - Implementation of system
 - Evaluation of system based on test document set and test query set
 - Refinement of model and re-evaluation

Retrieval Evaluation

- Assuming the system passes the functional analysis phase (ie satisfies the required functionality), performance evaluation will then take place
 - Time and space performance (as with other systems)
 - Similar issues to DBMS in terms of examining use and choice of architecture, index structures, interaction with OS, communication etc.
 - Retrieval performance evaluation
 - Since users queries are often vague and the set/list of retrieved documents not an exact answer, there needs to be some means of evaluating the precision of the answer
 - To do this, we need specialists to provide answers to the test queries in terms of the documents that they consider to be relevant and then compare this with results *returned by system in terms of two measures Recall and Precision*

Recall and Precision

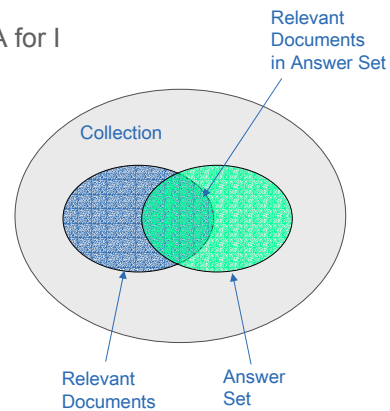
- Let I be an information request and R its set of relevant documents, where $|R|$ is the no. of docs in R
- Assume system generates answer set A for I

$$\text{Recall} = \frac{|R \cap A|}{|A|}$$

fraction of relevant documents retrieved

$$\text{Precision} = \frac{|R \cap A|}{|R|}$$

fraction of retrieved documents which are relevant



Recall and Precision ...

- As defined, we assume that all documents in the answer set are examined
- But users presented with documents in a ranked order and examine them in order
- Then recall and precision measures vary as the user proceeds with their examination of the answer set
- Thus proper evaluation requires plotting a precision versus recall curve

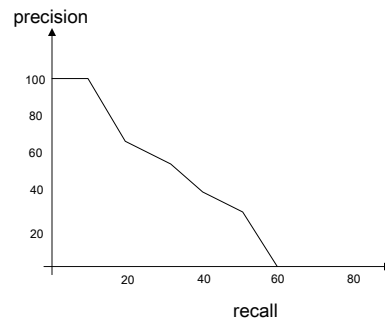
Plotting Precision against Recall

Example

For a query q
assume the
following
answer set

1. d_{123}
2. d_{84}
3. d_{56}
4. d_6
5. d_8
6. d_9
7. d_{511}
8. d_{129}
9. d_{187}
10. d_{25}
11. d_{38}
12. d_{48}
13. d_{250}
14. d_{113}
15. d_3

assume experts have
determined a set of 10 queries
which are relevant to the query



Alternative Measures

- A number of alternative measures have been proposed that attempt to combine recall and precision

Let $r(j)$ be recall and $P(j)$ precision for j -th document in ranking

Harmonic Mean
$$F(j) = \frac{2}{\frac{1}{r(j)} + \frac{1}{P(j)}}$$

*determination of max value
for F attempts to find
compromise between recall
and precision*

E-Measure
$$E(j) = 1 - \frac{1 + b^2}{\frac{b^2}{r(j)} + \frac{1}{P(j)}}$$

*allows the user to specify
whether they are more
interested in recall or
precision by specifying
parameter b*

Index Structures

- Special index structures are used to support the searching process
- Three main indexing techniques commonly used are:
 - [inverted files](#)
 - [suffix arrays](#)
 - [signature files](#)
- We will look at these in the next lectures when we also deal with the web and multimedia documents and how the construction and use of index structures is adapted in these cases

Reference Material

- Not so many basic text books on information retrieval are available as for databases. This is due to fact that the IR community is much smaller than the DB community and there are not so many courses on IR.
- The following book is recommended as covering both the basic concepts and also more specialised and recent topics and models.
 - *Modern Information Retrieval*
[Ricardo Baeza-Yates and Berthier Ribeiro-Neto](#)
pub. Addison Wesley, 1999

Next Lecture

Information Retrieval

Searching the Web



11 January 2005